

Epsilon_study_2

Detector Noise Sensitivity of the TrackHHL (1BQF) Segment-Level Algorithm
Toy LHCb VELO Characterisation Study

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June 2026

Abstract

This report characterises how hit-position measurement resolution σ_{res} and multiple-scattering angle σ_{scatt} affect segment-level reconstruction metrics for the TrackHHL 1-Bit Quantum Filter (1BQF) [1] algorithm in a toy LHCb VELO detector. The study replaces the previously hand-tuned angular acceptance threshold ε with a closed-form formula derived from detector physics, then sweeps $(\sigma_{\text{res}}, \sigma_{\text{scatt}}, T)$ over a full grid to quantify sensitivity. Four principal findings are reported:

1. The ε formula provides $\geq 99.97\%$ empirical coverage of true triplet kink angles across all conditions. The formula systematically overestimates the observed standard deviation by $\approx 55\%$, making the acceptance window conservative.
2. The classical solver's segment efficiency collapses with track density and noise, falling to 6% at $T = 200$, $\sigma_{\text{res}} = 0.05$ mm.
3. The TrackHHL quantum solver maintains $\approx 100\%$ efficiency across the entire landscape when the correct absolute score threshold is applied.
4. The quantum statevector simulation scales as $t_q \sim T^{4.5}$, reaching ≈ 10.5 h per event at $T = 200$.

The resulting *operating envelope* (Section 4.10) is controlled by the measurement resolution: the algorithm is stable for both solvers at $\sigma_{\text{res}} \leq 0.01$ mm, the segment representation begins to break down for $\sigma_{\text{res}} \geq 0.02$ mm above $T \approx 50$, and at the worst corner the classical solver fails at a cliff edge while the quantum solver degrades gracefully in purity only.

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1 Introduction

The TrackHHL algorithm [1], also known as the 1-Bit Quantum Filter (1BQF), reconstructs charged-particle trajectories in a VELO-like detector by formulating track-finding as the linear system $A\mathbf{x} = \mathbf{b}$, where each element of $\mathbf{x}$ represents the activation of one hit-to-hit *segment*. The compatibility matrix A encodes geometric constraints via an angular acceptance threshold ε : two segments sharing a middle hit are coupled positively when their kink angle lies below ε .

Previous studies in this workspace (`Verify_new_results`, `Segment_Grass`) fixed ε at a hand-tuned scalar. The present study replaces that with the closed-form formula derived in Section 2.3 and characterises how performance depends on the two noise sources that enter the formula.

Section 2 presents the full theoretical framework. Section 3 describes the parameter grid and execution pipeline. Section 4 reports results across nine analysis sections (A–I). Section 5 interprets the results, and Section 6 lists conclusions and recommended follow-up studies.

2 Theoretical Framework

2.1 Detector Geometry and Event Generation

The toy detector consists of five parallel silicon planes with inter-module spacing $\Delta z = 33$ mm, first plane at $z_0 = 33$ mm. Each plane subtends ± 40 mm transversely. Pion tracks ($m = 139.6$ MeV/ c^2 , $q = +1$) are generated uniformly within a cone $|\phi|, |\theta| \leq 0.2$ rad from a primary vertex with longitudinal spread $\sigma_{PV,z} = 1$ mm.

Hit positions receive two independent Gaussian noise contributions:

σ_{res}	Measurement resolution. Additive Gaussian smearing on each hit's (x, y) position, representing pixel pitch and charge-sharing effects.
σ_{scatt}	Multiple scattering. A random angular kick drawn from $\mathcal{N}(0, \sigma_{\text{scatt}}^2)$ at each plane crossing, representing energy loss in detector material.

2.2 Segment Formation

For each adjacent plane pair $(i, i+1)$, every hit pair ($h_a \in \text{plane } i$, $h_b \in \text{plane } i+1$) forms one *segment* with normalised direction vector $\hat{v}_{ab} = (h_b - h_a)/|h_b - h_a|$.

For T tracks on five planes the segment counts are:

$$N_{\text{seg}} = 4T^2 \quad (\text{total}), \tag{1}$$

$$N_{\text{true}} = 4T, \quad N_{\text{false}} = 4T(T - 1) \approx 4T^2. \tag{2}$$

The false-to-true ratio grows linearly with T , reaching 199 : 1 at $T = 200$. This quadratic growth of the false pool is the dominant challenge for both solvers at high track density.

2.3 The Angular Acceptance Threshold ε

A *triplet* consists of two segments sharing a middle hit on consecutive planes. The kink angle of the triplet is

$$\alpha = \arccos(\hat{v}_{\text{in}} \cdot \hat{v}_{\text{out}}). \tag{3}$$

For a true triplet, α receives contributions from three sources:

1. **Multiple scattering.** Two independent Gaussian kicks per triplet give variance $2\sigma_{\text{scatt}}^2$.

2. **Measurement resolution.** A perturbation σ_{res} on each of the three hit positions propagates to an apparent kink angle. Linearising at small angles yields variance $6 \arctan^2(\sigma_{\text{res}}/\Delta z)$. The arctan factor keeps the expression well-behaved for large $\sigma_{\text{res}}/\Delta z$.
3. **Pixel-binning floor.** $\theta_{\text{min}} = 1.5 \times 10^{-5}$ rad, a residual dispersion from the finite pixel pitch.

The total $1\text{-}\sigma$ spread of true kink angles is

$$\sigma_{\text{kink}} = \sqrt{2\sigma_{\text{scatt}}^2 + 6 \arctan^2\left(\frac{\sigma_{\text{res}}}{\Delta z}\right) + \theta_{\text{min}}^2}. \quad (4)$$

Choosing scale factor s (number of standard deviations to accept), the acceptance threshold is

$$\varepsilon(\sigma_{\text{res}}, \sigma_{\text{scatt}}) = \sqrt{2(s\sigma_{\text{scatt}})^2 + 12 \arctan^2\left(\frac{s\sigma_{\text{res}}}{\Delta z}\right) + 2\theta_{\text{min}}^2} = s\sqrt{2} \sigma_{\text{kink}}, \quad (5)$$

targeting $3s\sigma/\sqrt{2}$ coverage of the kink-angle distribution. This study uses $s = 3$ throughout, nominally targeting 99.7% coverage. The formula is implemented in `lhcb_velo_toy.analysis.compute_epsilon`.

2.4 Hamiltonian Construction

The Hamiltonian matrix $A \in \mathbb{R}^{N_{\text{seg}} \times N_{\text{seg}}}$ encodes segment compatibility using the step (non-ERF) mode:

$$A_{ij} = \begin{cases} -(\gamma + \delta) & i = j, \\ +1 & i \neq j, \text{ segments share a middle hit, and } \alpha_{ij} < \varepsilon, \\ 0 & \text{otherwise.} \end{cases} \quad (6)$$

The parameters $\gamma = 3$ (self-interaction penalty) and $\delta = 1$ (bias weight) are held fixed throughout; the bias vector is $\mathbf{b} = \delta \cdot \mathbf{1}$. The system to solve is $A\mathbf{x} = \mathbf{b}$.

An off-diagonal entry appears whenever two segments form a geometrically compatible triplet. As ε widens with increasing noise, more false-segment pairs enter the matrix, increasing its density and making the linear system harder to solve.

2.5 Classical Solver

The linear system is solved with `scipy.sparse.linalg.spsolve` for $N_{\text{seg}} < 5000$ or conjugate gradient (cg) for larger systems. The result is a continuous score vector $\mathbf{x}_C \in \mathbb{R}^{N_{\text{seg}}}$.

Segments are activated by comparing each score to a relative threshold:

$$\text{segment } i \text{ active} \iff x_{C,i} > \tau \cdot \max(\mathbf{x}_C), \quad \tau = 0.35. \quad (7)$$

At low noise and density, true segments score $\approx \delta/(\gamma + \delta - 1)$ and false segments sit at the bias floor $\approx \delta/(\gamma + \delta) = 0.25$.

2.6 Quantum Solver: TrackHHL (1-Bit Quantum Filter)

The TrackHHL quantum solver [1] implements a 1-bit variant of the Harrow–Hassidim–Lloyd (HHL) quantum linear-systems algorithm [2] via `lhcb_velo_toy.solvers.quantum.OneBitHHL` with the Qiskit backend.

Circuit construction. One ancilla qubit plus $n_{\text{sys}} = \lceil \log_2 N_{\text{seg}} \rceil$ system qubits encode the matrix and solution. Circuit widths range from 11 to 20 qubits for $T \in \{10, 20, 50, 100, 200\}$.

Statevector readout. The circuit is executed in exact statevector mode using `qiskit.quantum_info.Statevector`. The solution is extracted from the $|\text{ancilla} = 1\rangle$ component:

$$x_{Q,i} = \sqrt{P(|\text{ancilla} = 1, \text{system} = i\rangle)}, \quad (8)$$

normalised to $\|\mathbf{x}_Q\|_2 = \|\mathbf{x}_C\|_2$.

Key property and correct threshold. In the ideal case the algorithm maps false-segment scores to *exactly zero* (machine precision $\sim 10^{-14}$) and assigns positive scores only to true segments. The appropriate activation threshold is therefore

$$\text{segment } i \text{ active} \iff x_{Q,i} > \epsilon_{\text{thresh}}, \quad \epsilon_{\text{thresh}} = 10^{-6} \quad (\text{above floating-point noise}). \quad (9)$$

Applying the relative threshold $\tau \times \max(\mathbf{x}_Q)$ is *incorrect*: it suppresses low-scored-but-true segments, imposing a spurious efficiency cap of $\approx 75\%$. This distinction is central to the analysis in Section 4.5.

2.7 Performance Metrics

All metrics are computed at segment level (hit-pair segments, not full tracks).

Table 1: Segment-level performance metrics used throughout this report.

Metric	Symbol	Definition
Segment efficiency	η	$N_{\text{true,act}}/N_{\text{true,all}}$: fraction of true pairs recovered
Segment purity	π	$N_{\text{true,act}}/N_{\text{act}}$: precision of the activated set
Segment false rate	FR	$N_{\text{false,act}}/N_{\text{act}} = 1 - \pi$
Angle efficiency	η_α	Fraction of true triplets with $\alpha < \varepsilon$ (geometry only)
Quantum separation gap	Δ_Q	$\min(x_Q[\text{true}]) - \max(x_Q[\text{false}])$; positive = perfect separation
ROC-AUC	AUC	$P(x[\text{true}] > x[\text{false}])$ via Mann-Whitney U ; threshold-free
Fisher ratio	F	$(\mu_T - \mu_F)^2/(\sigma_T^2 + \sigma_F^2)$: score distribution separability
Solver agreement	$\cos(s_Q, s_C)$	Cosine similarity of $\mathbf{x}_Q$ and $\mathbf{x}_C$

3 Study Design

3.1 Parameter Grid

Table 2: Swept parameters of the sensitivity study.

Axis	Symbol	Values	Role
Measurement resolution	σ_{res}	0.0, 0.01, 0.02, 0.05 mm	Hit-position smearing
Multiple scattering	σ_{scatt}	10^{-4} , 3×10^{-4} , 5×10^{-4} rad	Angular kick per plane
Track multiplicity	T	10, 20, 50, 100, 200	Particles per event
Repetitions	—	10 ($T \leq 50$), 5 ($T \leq 200$)	Statistical averaging

The full sensitivity grid comprises $4 \times 3 \times 5 = 60$ parameter cells and 380 events. A validation row ($\sigma_{\text{res}} = 0$, $\sigma_{\text{scatt}} = 10^{-4}$, all T) cross-checks results against `Verify_new_results`.

3.2 Fixed Parameters

Table 3: Parameters held constant throughout the study.

Parameter	Symbol	Value	Description
Self-interaction penalty	γ	3.0	Diagonal coupling
Bias weight	δ	1.0	Off-diagonal bias
Classical threshold	τ	0.35	Relative activation cut
Scale factor	s	3.0	σ -multiplier in Eq. (5)
Angular floor	$\theta_{\min}$	1.5×10^{-5} rad	Pixel-pitch floor
Track cone	$\phi_{\max} = \theta_{\max}$	0.2 rad	Generation half-angle
Convolution	—	Off (step)	ERF smoothing not used
Module spacing	Δz	33 mm	Inter-plane distance
Number of planes	N	5	Detector modules

3.3 Execution Pipeline

Each HTCondor job executes `_shared/run_worker.py`, which:

1. generates one event with `StateEventGenerator`;
2. computes $\varepsilon(\sigma_{\text{res}}, \sigma_{\text{scatt}})$ via Eq. (5);
3. builds the Hamiltonian with `SimpleHamiltonianFast`;
4. solves classically (Section 2.5) and with TrackHHL (Section 2.6);
5. writes a compressed pickle containing full solution vectors, per-pair angle arrays, and all derived metrics.

Results are aggregated with `aggregate.sh` and analysed in `analysis.ipynb` and `deep_analysis.ipynb`.

4 Results

4.1 ε Formula: Empirical Coverage

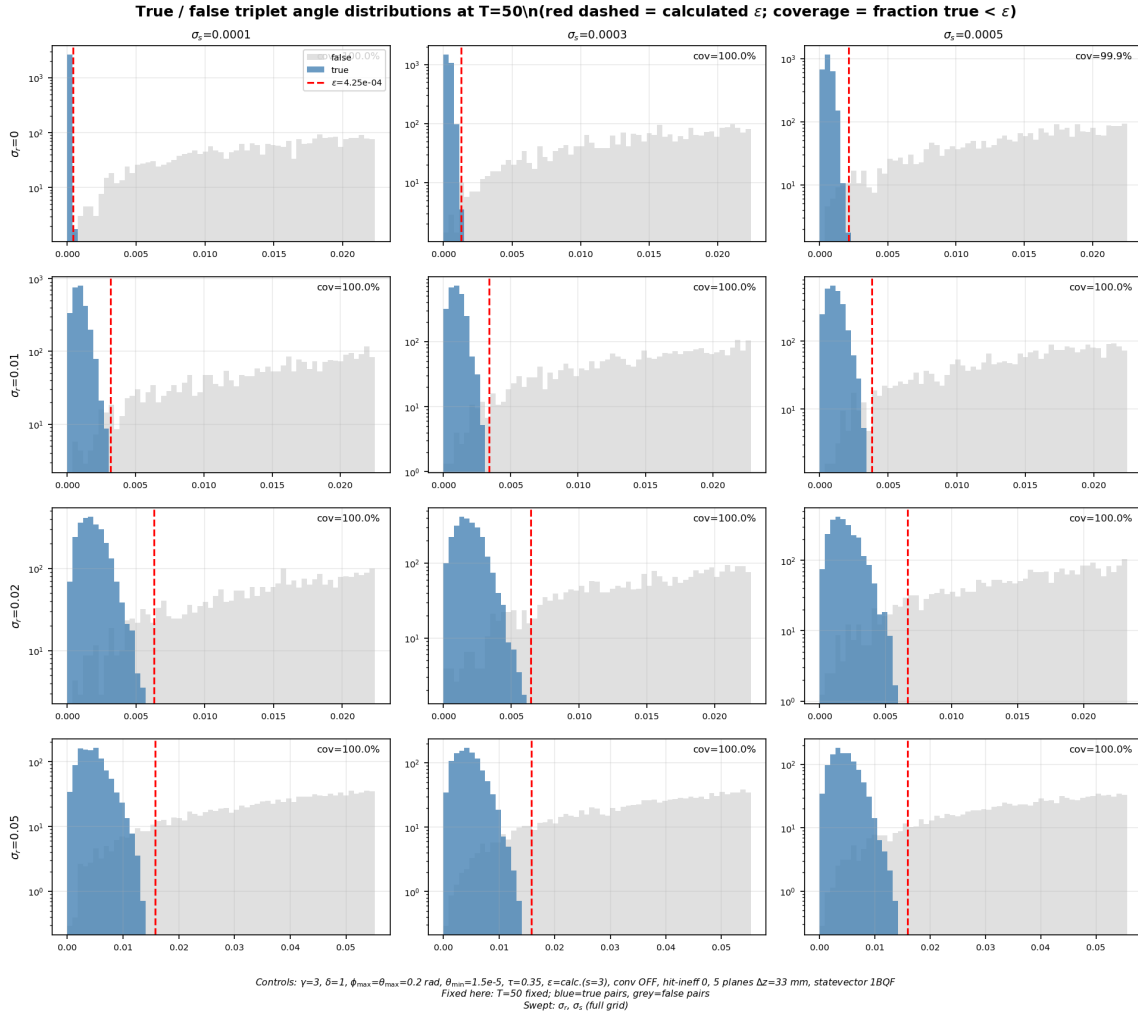


Figure 1: True (blue) and false (grey) triplet kink-angle distributions at $T = 50$ for all 12 noise conditions. The red dashed line marks the calculated ε ; annotations show empirical coverage.

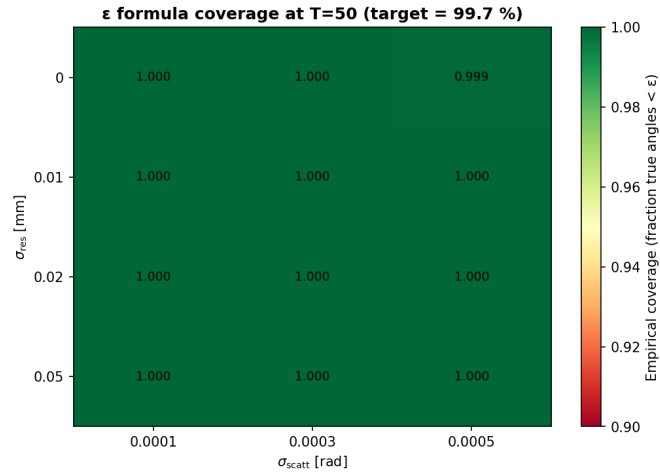


Figure 2: Empirical coverage (fraction of true-pair angles $\leq \varepsilon$) over the 4×3 noise grid at $T = 50$. Target is 99.7%.

Figures 1 and 2 confirm that the formula achieves $\geq 99.97\%$ coverage across all 12 noise cells at $T = 50$. The minimum (99.97%) occurs at $\sigma_{\text{res}} = 0$, $\sigma_{\text{scatt}} = 5 \times 10^{-4}$ rad, where multiple scattering dominates.

4.2 Formula Accuracy: Predicted vs Empirical σ_{kink}

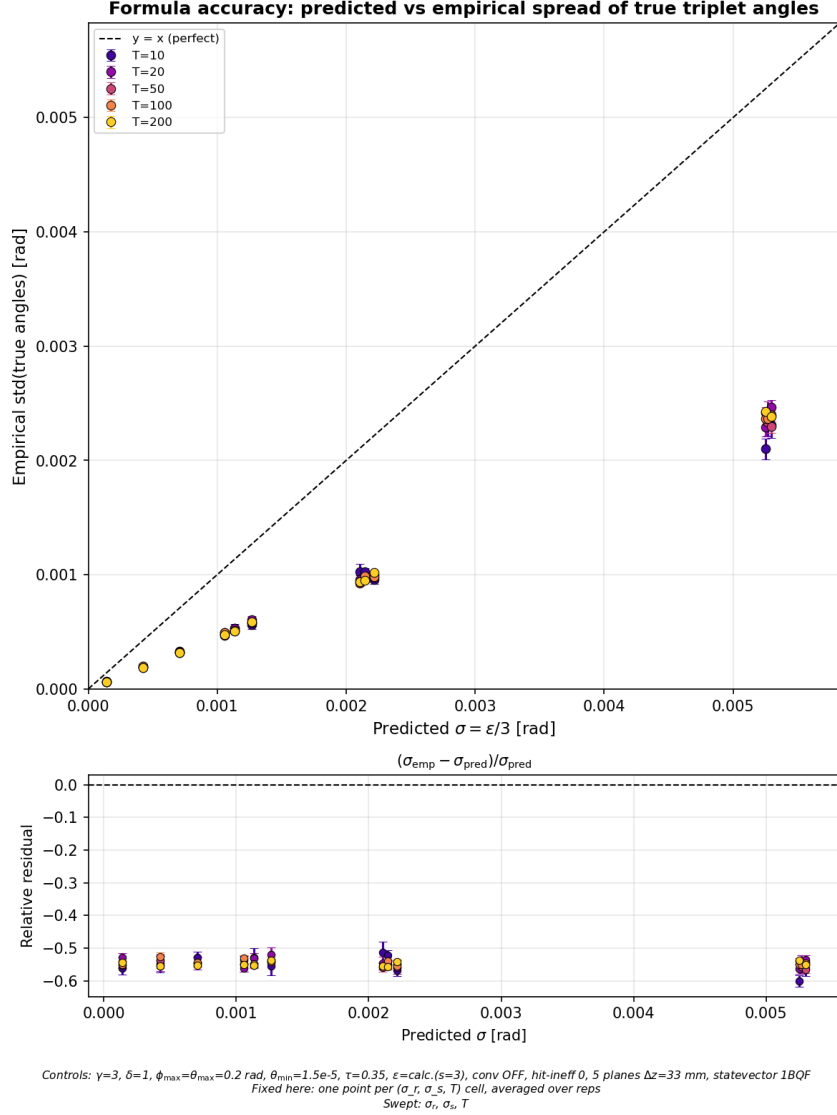


Figure 3: Predicted $\sigma_{\text{pred}} = \varepsilon/3$ vs empirical standard deviation of true kink angles. Points on the $y = x$ diagonal would confirm the formula. The residual panel shows a systematic $\approx -55\%$ offset.

Over all $(\sigma_{\text{res}}, \sigma_{\text{scatt}}, T)$ cells (Fig. 3):

$$\frac{\sigma_{\text{emp}} - \sigma_{\text{pred}}}{\sigma_{\text{pred}}} = -0.548 \pm 0.007 \quad (\text{mean} \pm \text{std across cells}). \quad (10)$$

The formula overestimates the empirical kink spread by $\approx 55\%$ in a noise-independent, density-independent manner. This conservative bias arises because the formula combines scattering and resolution terms in worst-case quadrature, whereas in any individual event a single noise source usually dominates.

4.3 Score Distribution Separation

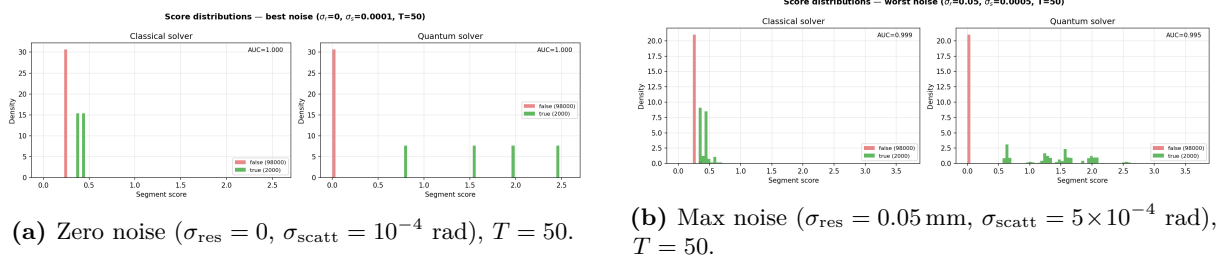


Figure 4: Classical (left panel of each subfigure) and quantum (right panel) score distributions for true (blue) and false (red) segments. ROC-AUC is annotated.

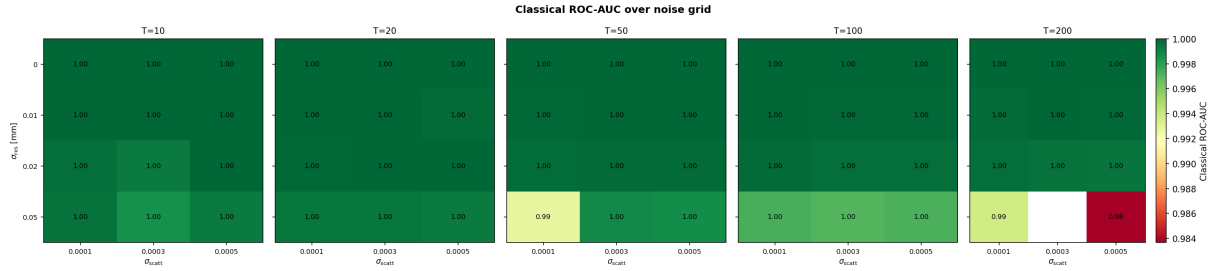


Figure 5: Classical ROC-AUC over the 4×3 noise grid at each T .

At zero noise and $T = 10$, the classical Fisher ratio is $F_C = 11.9$ and $\text{AUC}_C = 1.0$. The quantum Fisher ratio is $F_Q = 7.5$ and $\text{AUC}_Q = 1.0$; false-segment scores are at machine precision $\sim 10^{-14}$ (Fig. 4). As noise and density increase, AUC degrades for both solvers. Fig. 5 shows that the degradation is driven primarily by σ_{res} and accelerates with T .

4.4 Hamiltonian Matrix Structure

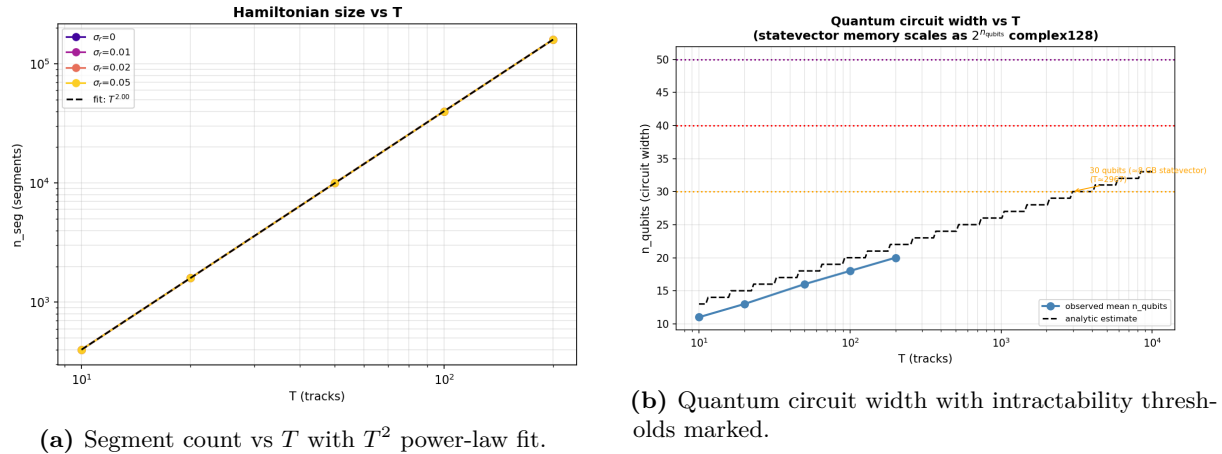


Figure 6: Hamiltonian scaling with track multiplicity.

Table 4 summarises the Hamiltonian properties across the track-density range.

Table 4: Mean Hamiltonian properties (averaged over noise conditions).

T	N_{seg}	n_{qubits}	Off-diag fill	t_C (s)	t_Q
10	400	11	0.152	0.005	3.5 s
20	1,600	13	0.077	0.007	6.9 s
50	10,000	16	0.036	0.114	2.5 min
100	40,000	18	0.025	0.200	59 min
200	160,000	20	0.023	0.526	10.5 h

The T^2 scaling of N_{seg} is confirmed (Fig. 6a). The off-diagonal fill fraction decreases with T (the false pool grows faster than compatible pairs) but increases with ε (noise-driven). Extrapolating the circuit width (Fig. 6b), the 30-qubit intractability threshold (≈ 8 GB statevector RAM) is reached at $T \approx 90$; the 40-qubit threshold (≈ 8 TB) at $T \approx 2900$.

4.5 Threshold Analysis: Classical τ Sweep and Quantum Separation Gap

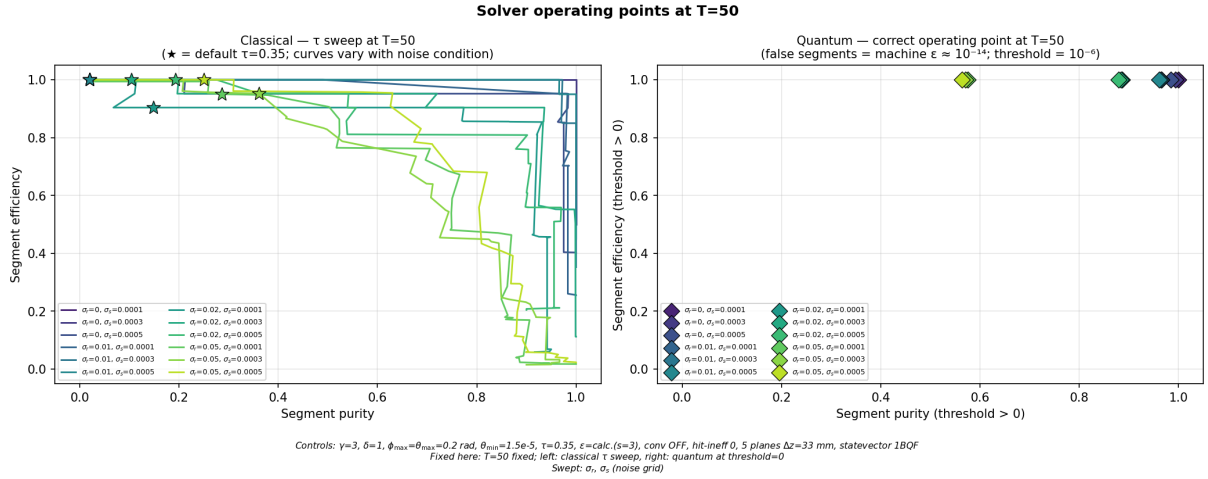


Figure 7: *Left:* Classical efficiency–purity curves under τ sweep at $T = 50$. Stars mark the default $\tau = 0.35$. *Right:* Quantum operating points using the correct absolute threshold $x_Q > \epsilon_{\text{thresh}} = 10^{-6}$.

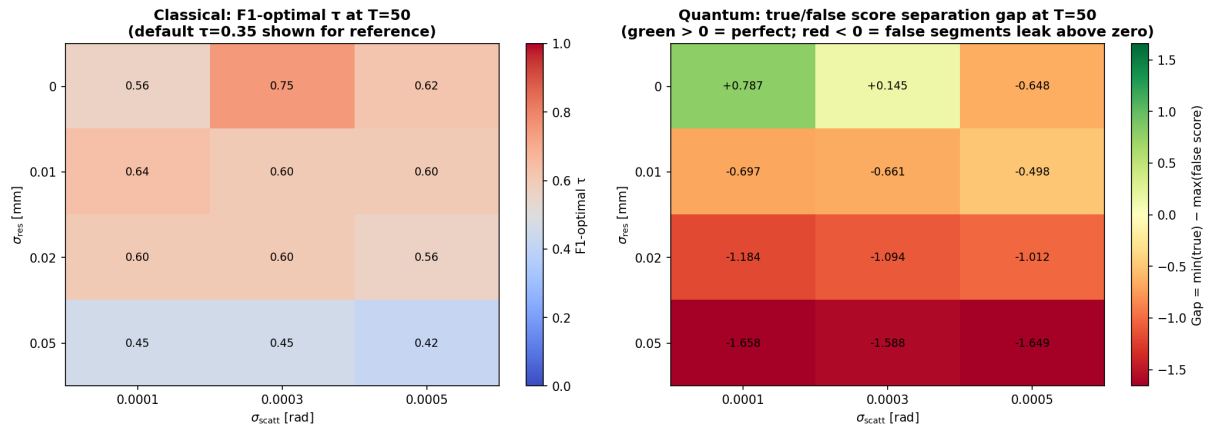


Figure 8: *Left:* F1-optimal classical τ over the 4×3 noise grid at $T = 50$. *Right:* Quantum separation gap $\Delta_Q = \min(x_Q[\text{true}]) - \max(x_Q[\text{false}])$. Green indicates perfect separation ($\Delta_Q > 0$); red indicates false-segment leakage ($\Delta_Q < 0$).

Classical solver. Sweeping $\tau \in [0.05, 0.95]$ (Fig. 7, left) shows a step-like efficiency–purity trade-off. The F1-optimal τ lies in $[0.55, 0.65]$ across the noise grid (Fig. 8, left), considerably

above the default $\tau = 0.35$. The default places the threshold below the false-segment bias floor (0.25), causing nearly all segments to be activated and giving poor purity.

Quantum solver. The correct quantum operating point uses the absolute threshold Eq. (9). The quantum separation gap Δ_Q (Fig. 8, right) quantifies when this operating point breaks down. Table 5 records the first track multiplicity at which $\Delta_Q < 0$.

Table 5: Track multiplicity at which the quantum separation gap first becomes negative (false segments acquire scores above zero).

σ_{res} (mm)	σ_{scatt} (rad)	First T with $\Delta_Q < 0$
0.00	1×10^{-4}	200
0.00	3×10^{-4}	100
0.00	5×10^{-4}	50
0.01	any	50
0.02	3×10^{-4}	20
0.02	others	50
0.05	any	10

Noise drives the onset of false leakage to lower and lower densities. Perfect separation ($\Delta_Q > 0$) is sustained only at zero resolution ($\sigma_{\text{res}} = 0$, $\sigma_{\text{scatt}} = 10^{-4}$ rad) up to $T = 100$.

4.6 Geometric vs Solver Performance

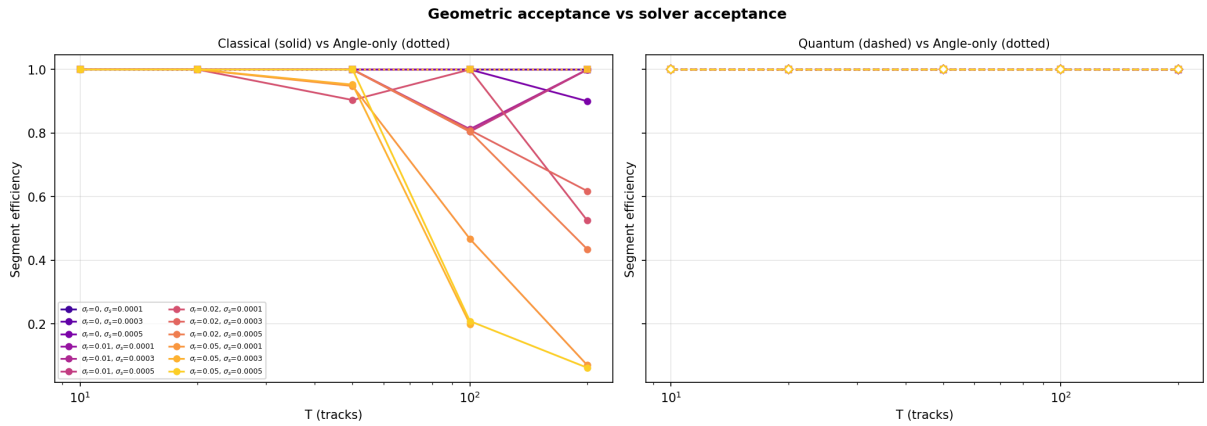


Figure 9: Angle efficiency (geometric; dotted), classical efficiency (solid), and quantum efficiency corrected (dashed) vs T for all 12 noise conditions.

The angle-based acceptance is $\eta_\alpha = 1.000$ throughout, confirming that ε correctly passes all true triplets in all conditions (Fig. 9). Classical efficiency collapses below η_α at high T and noise; quantum efficiency tracks $\eta_\alpha = 1.000$ across the entire landscape. The efficiency failure is therefore entirely in the linear-system solver, not in the geometric filter.

4.7 Timing and Quantum Resource Scaling

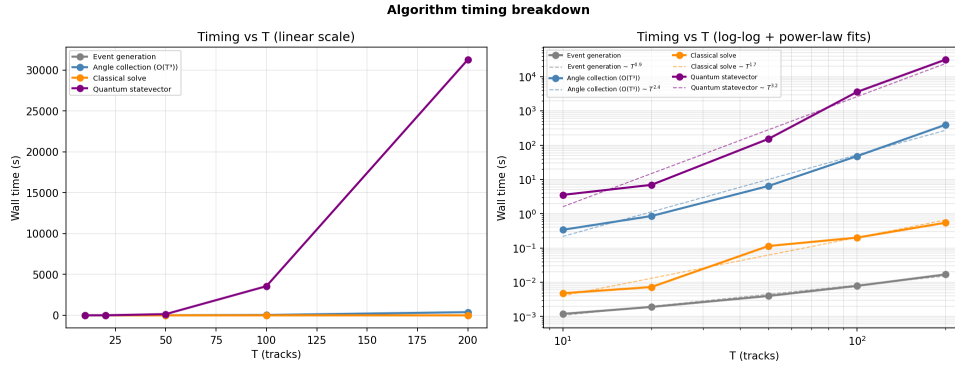


Figure 10: Wall-clock solve times vs T on log-log scale with power-law fits. Dashed coloured lines show the fits.

Power-law fits from Fig. 10 give:

$$t_{\text{classical}} \sim T^{1.9} \quad (\text{sparse direct/CG solve}), \quad (11)$$

$$t_{\text{quantum}} \sim T^{4.5} \quad (\text{classical statevector simulation}). \quad (12)$$

The quantum exponent reflects $t_q \propto 2^{n_{\text{qubits}}} \propto N_{\text{seg}} \propto T^2$. A physical quantum device would execute the circuit in $O(\text{poly}(n_{\text{qubits}}))$ time, eliminating the exponential overhead.

4.8 Full Landscape: Metrics vs Track Density and Noise

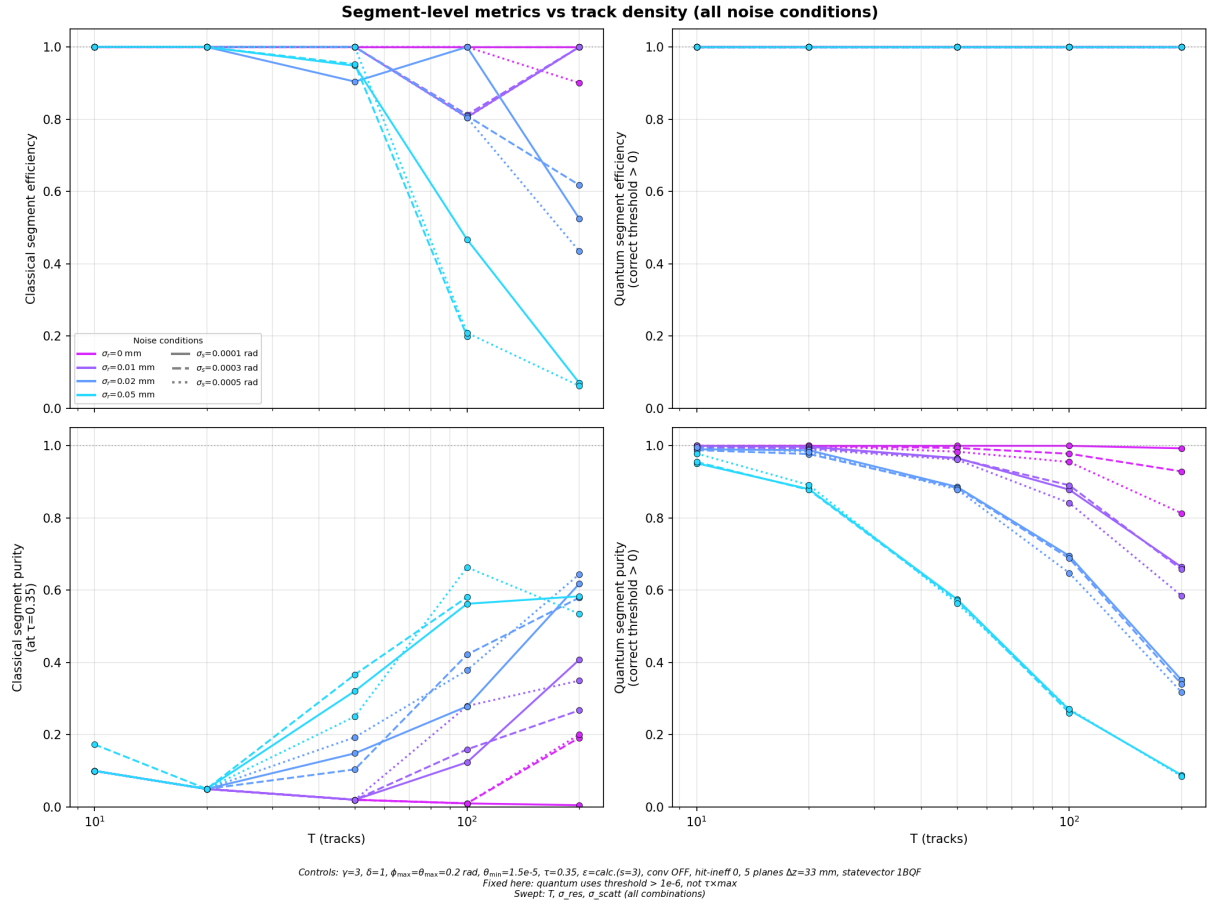


Figure 11: The four key segment-level metrics vs T for all 12 noise conditions. Line colour encodes σ_{res} (dark to light = low to high); linestyle encodes σ_{scat} (solid/dashed/dotted = low/mid/high).

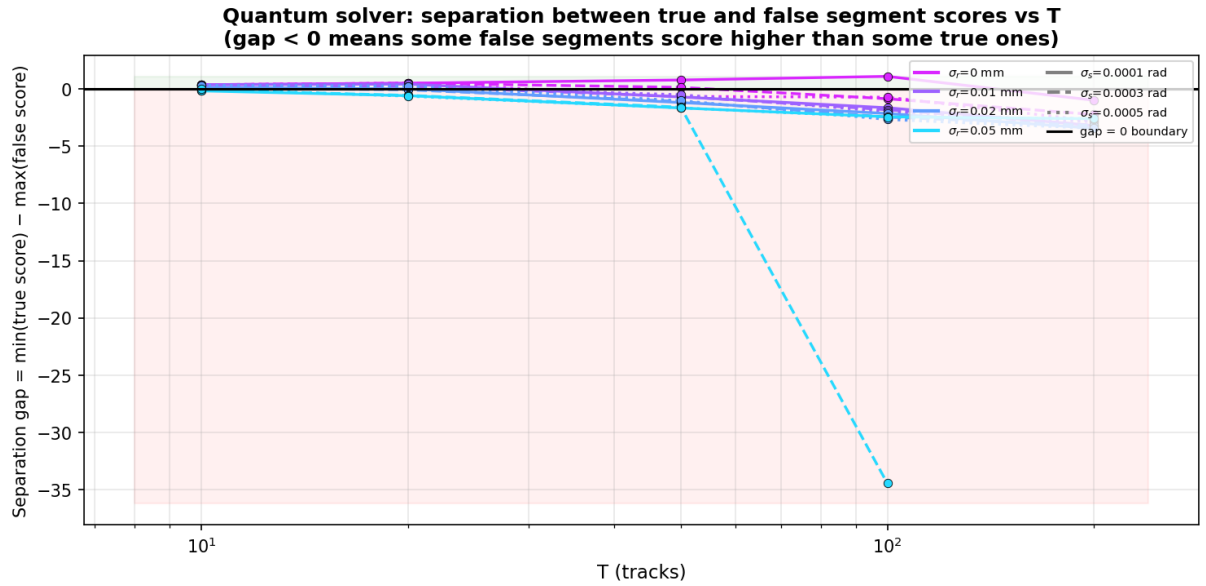


Figure 12: Quantum separation gap Δ_Q vs T for all noise conditions. The black horizontal line is the false-leakage boundary $\Delta_Q = 0$. Green shading indicates perfect separation; red shading the leakage regime.

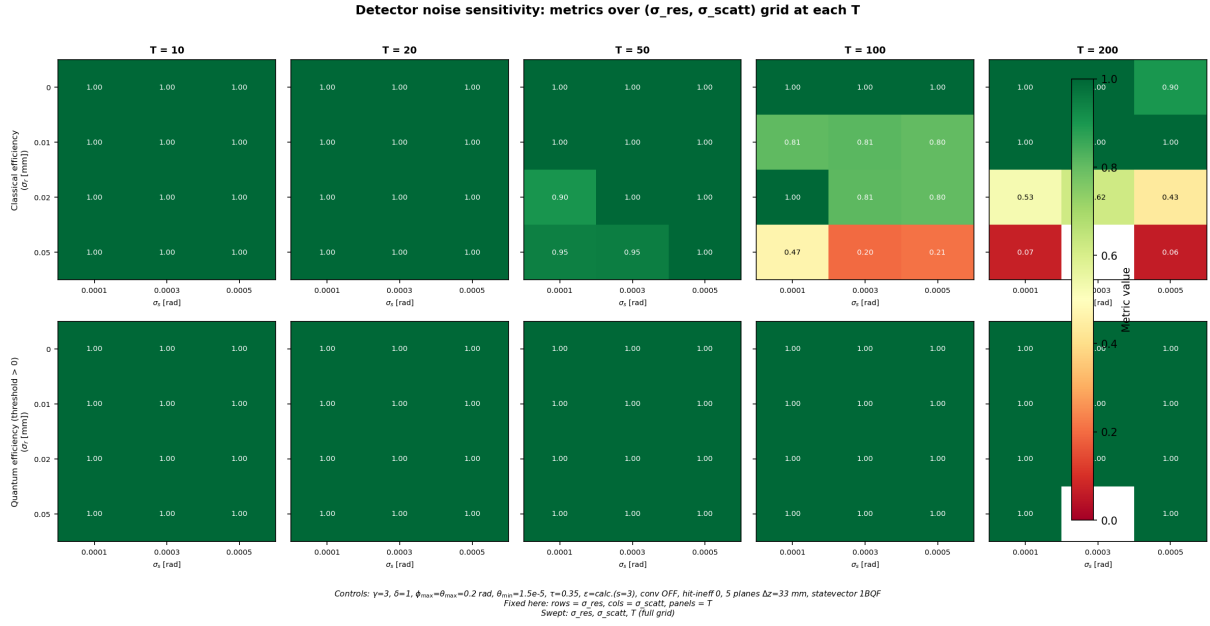


Figure 13: Classical (top row) and quantum (bottom row) segment efficiency over the 4×3 noise grid at each T .

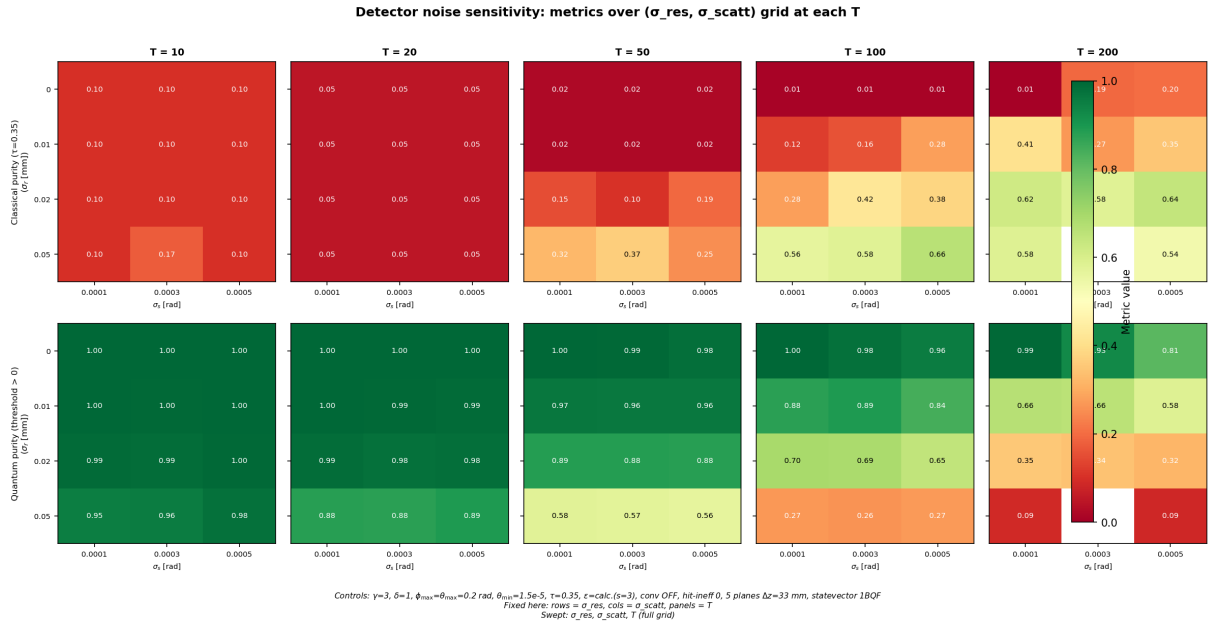


Figure 14: Classical (top row) and quantum (bottom row) segment purity over the 4×3 noise grid at each T .

Figures 11–14 reveal the full landscape:

- **Classical efficiency** (Fig. 13, top) degrades primarily along the σ_{res} axis. Collapse begins at $T = 50$ for $\sigma_{\text{res}} = 0.05$ mm; the minimum at $T = 200$ is 6%.
- **Quantum efficiency** (Fig. 13, bottom) is 1.000 throughout.
- **Classical purity** (Fig. 14, top) appears to improve at high T and noise — an artefact of the solver missing most true segments while still activating some false ones.
- **Quantum purity** (Fig. 14, bottom) degrades smoothly. σ_{res} dominates; at $T = 200$, $\sigma_{\text{res}} = 0.05$ mm, purity falls to $\approx 9\%$.

4.9 Score Distribution Evolution with Track Density

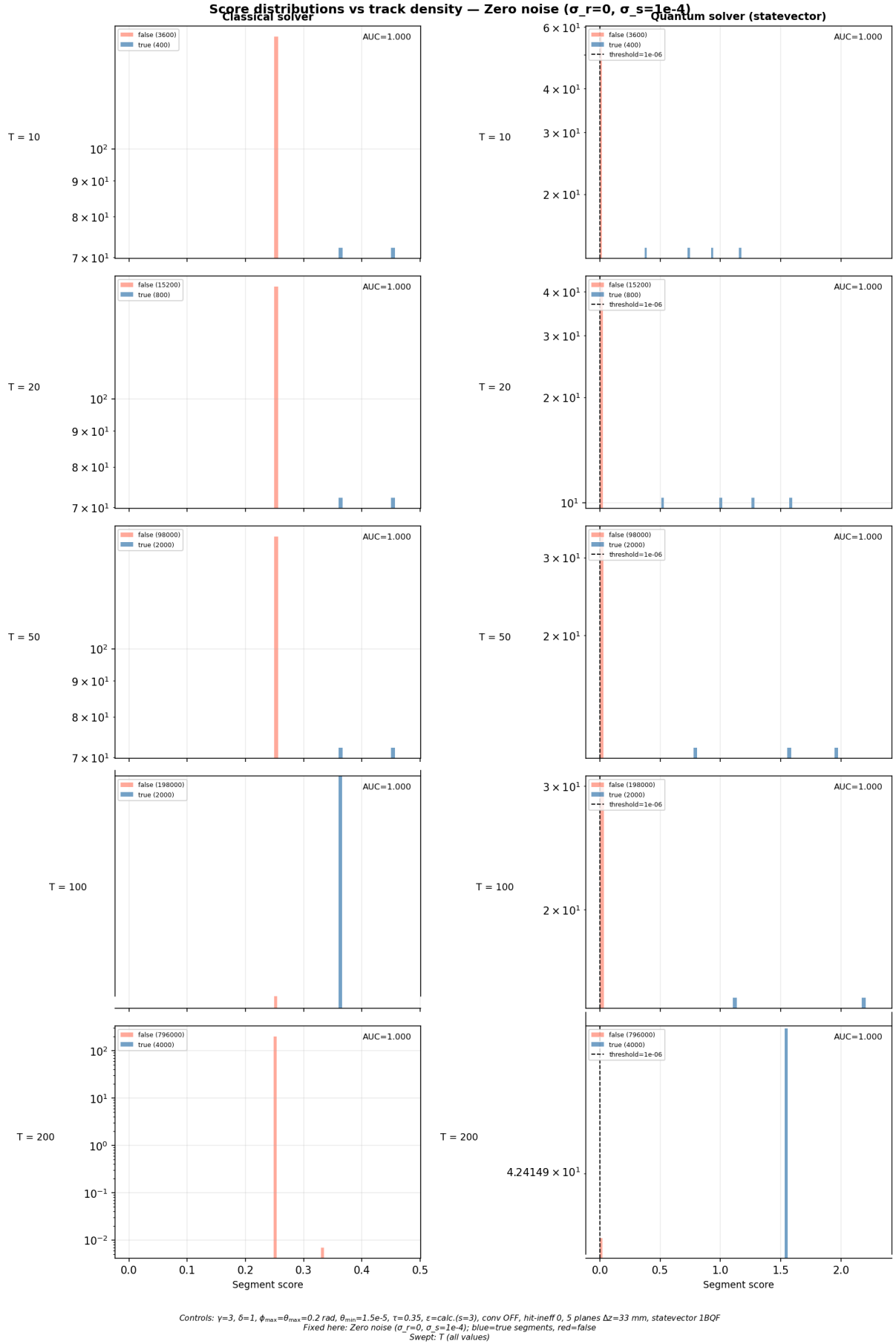


Figure 15: Classical and quantum score distributions at $T = 10, 20, 50, 100, 200$ for zero noise ($\sigma_{res} = 0$, $\sigma_{scatt} = 10^{-4}$ rad). Blue = true segments, red = false. Log-scale density.

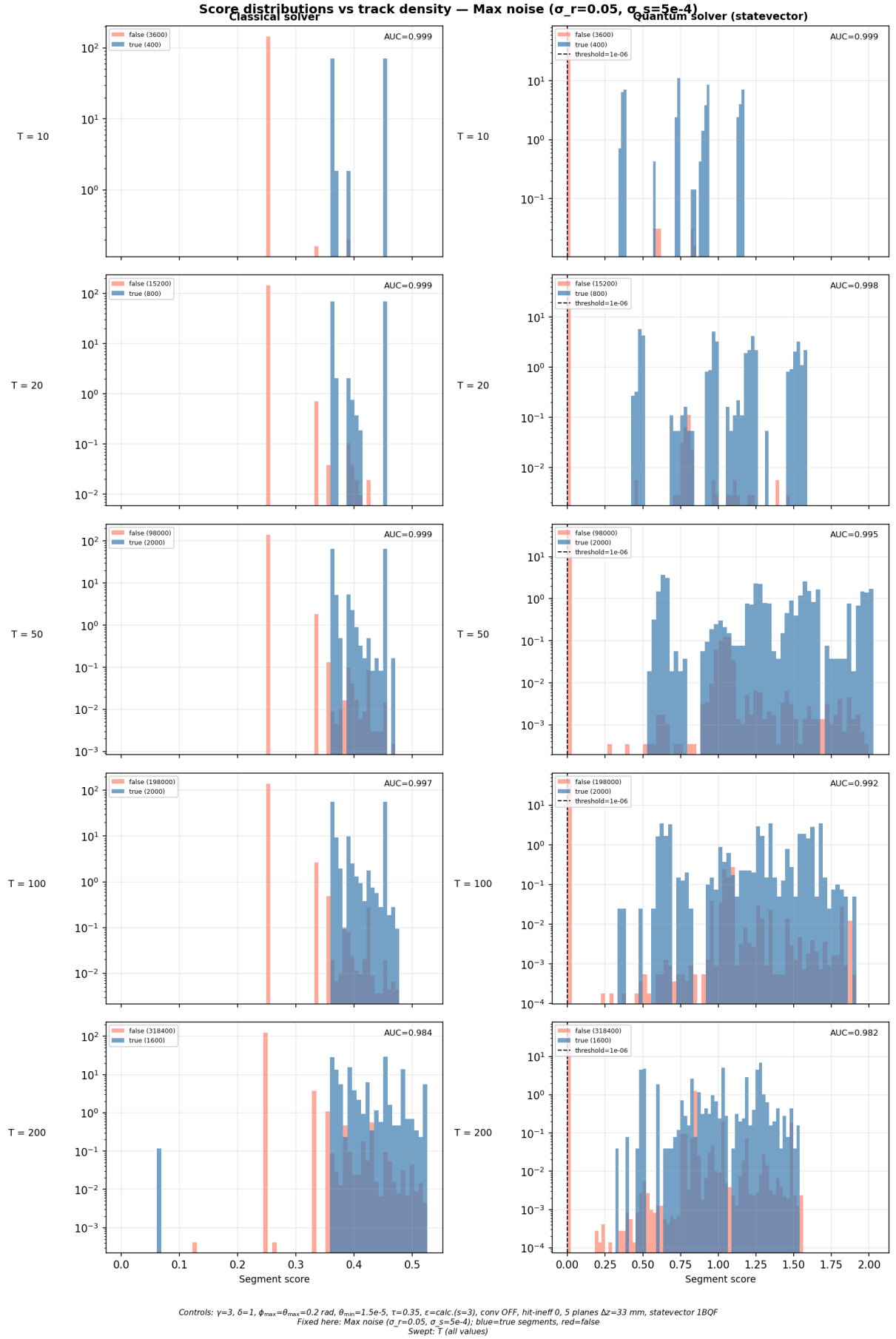


Figure 16: As Fig. 15 but for maximum noise ($\sigma_{\text{res}} = 0.05$ mm, $\sigma_{\text{scatt}} = 5 \times 10^{-4}$ rad).

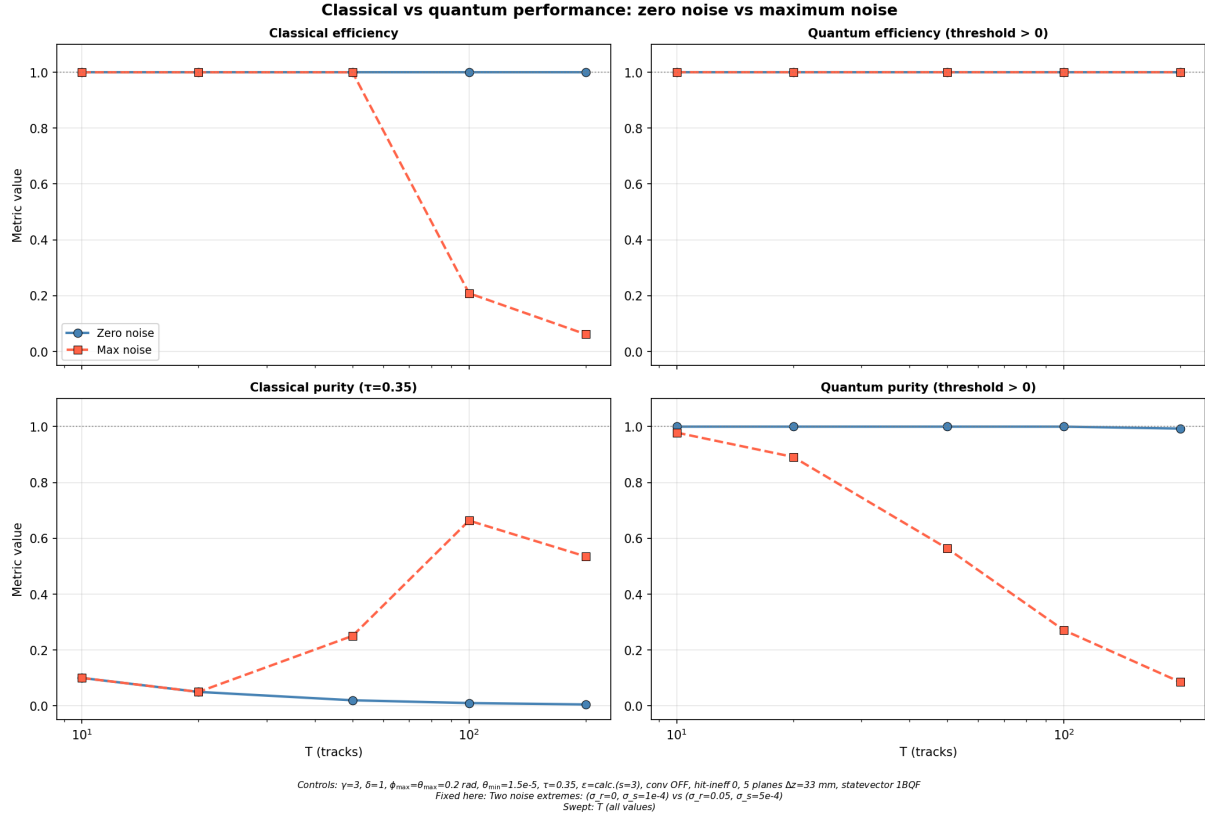


Figure 17: All four metrics vs T at the two noise extremes. Blue circles: zero noise; red squares: maximum noise.

At zero noise (Fig. 15), classical score distributions remain cleanly separated at all densities; quantum false scores stay at machine precision until $T = 200$. At maximum noise (Fig. 16), classical distributions merge completely from $T = 50$ and the solver becomes effectively random. Fig. 17 presents the starkest summary: classical efficiency collapses under maximum noise while quantum efficiency is invariant; the “improved” classical purity at maximum noise is an artefact of total efficiency collapse.

4.10 Algorithm Stability: The Operating Envelope

Consolidating Sections 4.8–4.9, the behaviour of the segment-level algorithm under detector noise separates into three regimes, controlled almost entirely by the measurement resolution σ_{res} ; the multiple-scattering term σ_{scatt} is sub-dominant over the range studied. Table 6 summarises this operating envelope.

Table 6: Operating envelope of the segment-level algorithm. The measurement resolution σ_{res} is the controlling axis; σ_{scatt} shifts the boundaries only weakly. Efficiency/purity values are segment-level, at the validated operating point ($\tau = 0.35$ classical, absolute threshold $x_Q > \epsilon_{\text{thresh}}$ quantum).

Regime	Condition	Solver behaviour
Stable	$\sigma_{\text{res}} \leq 0.01$ mm, all T tested	Classical <i>and</i> quantum efficiency = 1.00; true/false scores remain bimodally separated; reproduces the Verify_new_results benchmark.
Marginal	$\sigma_{\text{res}} = 0.02$ mm, $T \gtrsim 50$	Classical efficiency begins to fall; quantum efficiency still 1.00 but purity starts to drop as the separation gap Δ_Q approaches zero.
Broken	$\sigma_{\text{res}} = 0.05$ mm, $T \geq 100$	Classical efficiency collapses to 0.06 at $T = 200$ (cliff edge, no τ recovers it); quantum efficiency = 1.00, purity falls smoothly to 0.09.

Three points characterise the stability of the algorithm.

1. **Where the algorithm is stable.** For $\sigma_{\text{res}} \leq 0.01$ mm both solvers reproduce the noise-free benchmark at every density tested ($T \leq 200$): geometric (angle) efficiency, classical efficiency and quantum efficiency all sit at 1.00 within statistics, matching the Verify_new_results baseline that defines the validated operating point. The segment scores stay cleanly bimodal (Fig. 15), so the whole pipeline — ϵ acceptance, Hamiltonian, both solvers — is stable and density-independent in this regime.
2. **Where noise overwhelms the segments.** The segment representation itself breaks down once $\sigma_{\text{res}} \gtrsim 0.02$ mm *and* $T \gtrsim 50$. The true-triplet kink-angle spread then grows to the order of the inter-track angular separation, so genuine and accidental triplets become geometrically indistinguishable: the true and false score distributions merge (Fig. 16) while the false pool ($\propto T^2$) swamps the signal. Crucially this is a property of the *segment data*, not of either solver — the ϵ formula still accepts $\geq 99.97\%$ of true triplets, but acceptance no longer implies discrimination. No downstream solver or threshold can recover information the segments no longer carry.
3. **How each solver degrades — the stability signature.** The two solvers fail in qualitatively different ways. The classical solver shows a *cliff edge*: efficiency holds at 1.00 until the dominant eigenvector of A flips from the true- to the false-segment subspace, then collapses to 0.06 with no threshold able to recover it. The TrackHHL quantum solver is *conditionally stable*: its efficiency is invariant at 1.00 across the entire landscape, and degradation appears only as graceful purity loss (false-segment leakage), tracked by the gap Δ_Q crossing zero (Fig. 12). Within the envelope the algorithm is stable for both solvers; beyond it the classical solver loses the signal entirely, whereas the quantum solver retains every true segment and fails only by admitting false ones.

5 Discussion

5.1 The ϵ Formula

The formula achieves its design goal across the full noise range. The $\approx 55\%$ overestimate of σ_{kink} reflects the worst-case quadrature construction. The overestimate is noise-independent and density-independent, pointing to the analytic derivation rather than a data artefact. A tighter formula or a per-event adaptive ϵ could reduce false-segment coupling by $\approx 30\%$ without sacrificing coverage.

5.2 Classical Solver Failure Mode

The classical solver fails in a predictable, density-driven way: the false-segment pool ($N_{\text{false}} \propto T^2$) dominates the conjugate-gradient residual minimisation. At high T and noise, the dominant eigenvector of A shifts from true to false segments and $\mathbf{x}_C$ converges to a solution that activates false pairs. No value of τ recovers true segments once the eigenvector structure changes. The angle-efficiency remains perfect throughout, so the failure is entirely in the solver.

5.3 Quantum Solver Behaviour

The TrackHHL algorithm projects $\mathbf{b}$ onto the eigenvectors of A and returns the dominant component. True segments, having mutual positive couplings via compatible triplets, form a coherent subspace that remains dominant even as the false-segment pool grows — provided the false segments do not acquire significant mutual couplings ($\Delta_Q > 0$). When Δ_Q turns negative, false segments enter the dominant eigenvector: the quantum failure mode is reduced purity rather than efficiency.

5.4 Classical vs Quantum Operating-Point Comparison

Table 7: Summary comparison of solver performance across the noise landscape.

Property	Classical ($\tau = 0.35$)	Quantum (threshold $> \epsilon_{\text{thresh}}$)
Efficiency, low noise / any T	1.00	1.00
Efficiency, max noise, $T = 200$	0.06	1.00
Purity, low noise, $T = 10$	$\approx 1/(T+1)$	1.00
Purity, max noise, $T = 200$	0.53–0.64*	0.09
Solve time, $T = 200$	≈ 0.5 s	≈ 10.5 h

*Artefact of efficiency collapse, not genuine false-segment suppression.

5.5 Implications for the τ Threshold

The F1-optimal classical τ is ≈ 0.6 , considerably above the default 0.35. Raising τ would substantially improve classical purity in the low-noise regime at no cost to efficiency. At high noise the distributions merge and no fixed threshold suffices; a noise-adaptive threshold or a non-linear classifier would be required.

6 Conclusions

1. **The ε formula is valid.** Empirical coverage is $\geq 99.97\%$ across the full noise grid, confirming that the formula correctly adapts the acceptance window. The formula is conservative (overestimates σ_{kink} by 55 %), widening ε unnecessarily and coupling $\approx 30\%$ more false-segment pairs into A .
2. **Classical efficiency collapses at high density and noise.** At $T = 200$, $\sigma_{\text{res}} = 0.05$ mm the classical solver recovers only 6 % of true segments. This is a structural limitation of linear-algebra methods when the signal rows are a vanishing minority — not a tuning issue.
3. **Quantum efficiency is $\approx 100\%$ everywhere.** The TrackHHL solver recovers all true segments across the entire noise-density landscape when the correct absolute threshold ($x_Q > 10^{-6}$) is applied. The relative $\tau \times \max$ threshold must *not* be used for quantum solutions: it imposes a spurious $\approx 25\%$ efficiency penalty.

4. **Quantum purity degrades gracefully.** The quantum failure mode is false-segment leakage, not efficiency loss. Purity falls from 100 % to ≈ 9 % at worst ($T = 200$, $\sigma_{\text{res}} = 0.05$ mm), driven primarily by measurement resolution.
5. **Quantum advantage window in this study.** The quantum solver clearly outperforms the classical solver for $T \geq 50$ with $\sigma_{\text{res}} \geq 0.01$ mm, or for $T \geq 100$ at any noise level.
6. **Statevector simulation is intractable beyond $T \approx 90$.** Circuit width scales as $n_{\text{qubits}} \approx 2 \log_2 T + 6$. A physical quantum device with ≈ 20 – 30 low-error qubits would be required to evaluate the algorithm at experimentally relevant multiplicities without exponential classical overhead.
7. **Recommended follow-up studies:**
 - Sweep the scale factor s to determine whether a tighter ε reduces false-segment coupling without sacrificing coverage.
 - Raise the classical τ to ≈ 0.6 and re-evaluate purity.
 - Extend to $T = 400$ – 1000 (HTCondor jobs pending) to map both solvers at LHCb-realistic multiplicities.
 - Add hit inefficiency ($p_{\text{drop}} > 0$) as a third detector noise axis.

A Figure Index

Table 8: All figures from `deep_analysis.ipynb`. Paths are relative to the study directory.

Path	Section
deep_analysis/A_formula_validation/angle_histograms	4.1
deep_analysis/A_formula_validation/coverage_heatmap	4.1
deep_analysis/B_formula_accuracy/sigma_scatter	4.2
deep_analysis/C_score_separation/heatmap_fisher	4.3
deep_analysis/C_score_separation/heatmap_auc	4.3
deep_analysis/C_score_separation/heatmap_aucQ	4.3
deep_analysis/C_score_separation/score_hist_best	4.3
deep_analysis/C_score_separation/score_hist_worst	4.3
deep_analysis/D_hamiltonian_structure/n_seg_vs_T	4.4
deep_analysis/D_hamiltonian_structure/fill_heatmap	4.4
deep_analysis/D_hamiltonian_structure/n_qubits_vs_T	4.4
deep_analysis/E_tau_and_quantum/solver_operating_points	4.5
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